PROJECT REPORT

SAKSHI Constructions - Building Construction Business Dashboard

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Role: Aspiring Data Analyst

1. Title: Building Construction Business Intelligence Dashboard using Power BI

2. Abstract:

This project presents a complete Business Intelligence solution for the construction industry. The SAKSHI Constructions Dashboard was developed to overcome the challenges of fragmented data and lack of real-time monitoring. By integrating data from four different operational tables (Cost, Material, Project, Task), an interactive dashboard with 6 pages was created. The dashboard provides key insights into financial performance, material consumption, and task efficiency, enabling data-driven decision making.

3. Objective of the Project:

- To create a centralized system for tracking all construction projects.
- To analyze Budget vs. Actual Cost to control expenses.
- To monitor Material-wise profit and cost.To track Task Completion Rate (Current Avg: 78.08%) and identify delays.
- To visualize City-wise and Manager-wise project distribution.

4. Tools & Technology Used:

- Power BI Desktop: For dashboard creation and visualization.
- Power Query: For data cleaning, removing duplicates, and data transformation.
- DAX (Data Analysis Expressions): For creating 14+ business KPIs and measures.
- Data Modeling: Star Schema design for optimized data relationship.

5. Data Description:

The project uses 4 interconnected tables:

1. CostTable (Fact Table): BUDGET, TOTAL_COST, REVENUE, PROFIT, LABOR_COST, WORKERS
2. ProjectTable (Dimension): PROJECT_ID, PROJECT_NAME, CITY, PROJECT_STATUS, MANAGER, START_DATE, END_DATE
3. MaterialTable (Dimension): MATERIAL_NAME, MATERIAL_SUPPLIER, COST, QUANTITY
4. TaskTable (Dimension): TASK_ID, TASK_NAME, TASK_STATUS, TASK_COMPLETION_%

6. Methodology & DAX Measures:

A Star Schema model was built connecting all dimension tables to the Cost fact table. Following DAX measures were created:

- Total Budget = SUM(CostTable[BUDGET])
- Total Cost = SUM(CostTable[TOTAL_COST])
- Total Revenue = SUM(CostTable[REVENUE])
- Total Profit = SUM(CostTable[PROFIT])
- Total Workers = SUM(CostTable[WORKERS])
- Avg Task Completion = AVERAGE(TaskTable[TASK_COMPLETION_%])

7. Dashboard Development (6 Pages):

Page 1 - Home Dashboard: Navigation and main KPIs - Revenue 4Bn, Budget 2Bn, Workers 28K, Profit 2Bn.

Page 2 - Project Overview: Donut chart for Projects by City, Bar chart for Projects by Manager.

Page 3 - Financial Analysis: Clustered column chart for Budget vs Cost, Line chart for Month-wise Cost Trend.

Page 4 - Material Management: Pie chart for Profit by Material (9 materials, 11.11% each), Bar chart for Cost by Project.

Page 5 - Task Monitoring: KPI cards, Bar chart for Completion % by Task, Donut for Task Status (60.2% In Progress).

Page 6 - Performance Summary: Overall summary and business insights.

8. Key Findings & Insights:

- Company is highly profitable - Revenue (4Bn) is double the Cost (2Bn).
- 60.2% tasks are still In Progress, which needs management attention to avoid delays.
- Material cost is well-controlled at 185M total, with all materials contributing equally to profit.
- Top 3 cities with most projects are Houston, New York and Chicago.

9. Conclusion:

The SAKSHI Construction Dashboard successfully provides a 360-degree view of the business. It helps management to control costs, optimize material usage, and track project progress in real-time. This project demonstrates practical skills in Data Cleaning, Data Modeling, DAX, and Business Storytelling through Power BI.

10. Future Scope:

- Integration of live SQL database.Adding predictive analysis for project delay.
- Creating alerts for budget overruns.

THANK YOU